

THINKTAC ASSIGNMENT ANSWERS

Part 2 – Written Explanation

The application was structured using Angular's component and service architecture to maintain a clear separation between presentation logic and data access. UI components are responsible only for displaying information and handling user interactions, while dedicated services communicate with Firebase Authentication and Firestore. This approach keeps the code modular, reusable, and easier to maintain as the application grows. TypeScript strict mode was enabled to improve type safety, reduce runtime errors, and make the codebase more reliable during development.

Firestore was selected because it provides real-time synchronization, which is ideal for displaying live enrolment counts and available seats without requiring manual refreshes. Firebase Authentication was used to implement simple role-based access for coordinators and learners while minimizing backend complexity. Angular's reactive programming model with Observables integrates naturally with Firestore's real-time data streams, making it straightforward to keep the dashboard synchronized with database updates.

If this application were deployed for approximately 10,000 daily users, several improvements would be introduced. Firestore security rules would be strengthened with role-based authorization to ensure coordinators can modify courses while learners can only enrol. Frequently accessed dashboard data would be optimized using Firestore indexes and pagination to reduce read operations. Cloud Functions would be used to perform server-side validation for enrolments, preventing race conditions when multiple learners attempt to register simultaneously. Logging, monitoring, and automated testing would also be added to improve reliability and simplify maintenance.

Due to the limited time available, advanced notification features were intentionally omitted. Given additional time, I would implement automated email notifications using Firebase Cloud Functions triggered after successful enrolments. Learners would receive confirmation emails, while coordinators would be notified when courses become full or when significant changes are made to course details. This would improve communication without increasing manual effort.

Beyond the required technologies, I primarily relied on Angular's built-in features and AngularFire to integrate Firebase services with Angular. AngularFire simplifies authentication, Firestore operations, and real-time data synchronization while following Angular best practices. No unnecessary third-party libraries were introduced, keeping the project lightweight, maintainable, and aligned with the assignment requirements.